Inputs of Equivariant Infinite Loop Space Machines

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1 Abstract

Symmetric monoidal categories are central studies in modern algebraic topology, for instance K-theory relies on constructing spectra with suitable homotopy groups out of symmetric monoidal categories using the infinite loop space machine. It's known that the two inputs - pesudoalgebra over an operad $\mathscr{P}$, or suitable categories described by based finite sets $\mathscr{F}$ - are equivalent description for symmetric monoidal categories.

Within equivariant K-theory, similar machineries have been established, where the input for constructing G-spectra are described by pseudoalgebra over an operad $\mathscr{P}_G$ as G-categories, or categories described by based finite G-sets $\mathscr{F}_G$. The equivalence of the two within equivariant case is a natural question to ask, and associates to the question of "what is symmetric monoidal G-categories".

We'll provide an exposition on the two inputs by collecting the axioms scattered (or implicitly assumed) in varioius literatures, while describe some equivariance properties of the two inputs.

2 Introduction (ND)

3 Categorical Preliminaries

3.1 2-Categories

Definition 3.1. A category $\mathscr{C}$ is a *2-category*, if for each object $X, Y \in \mathscr{C}$ (called *0-cells*), the homset $\mathscr{C}(X,Y)$ is itself a category (called the *hom-category*). Also, for every 0-cells $X, Y, Z \in \mathscr{C}$, the *composition* $\circ : \mathscr{C}(Y,Z) \times \mathscr{C}(X,Y) \to \mathscr{C}(X,Z)$ is a functor.

Every morphism $f \in \mathscr{C}(X,Y)$ is called a *1-cell*, while given any 1-cells $f, g \in \mathscr{C}(X,Y)$, a morphism $\alpha : f \Rightarrow g$ in the hom-category $\mathscr{C}(X,Y)$ is called a *2-cell*.

Here, $\circ$ must be strictly unital and associative.

Remark 3.2. There is a more general notion of a *bicategory*, which relaxes the composition functors to be associative and unital up to coherent natural isomorphisms, and in some literature this is called 2-categories. Here we'll take the strict associativity and unitality as axioms for 2-categories. For more general bicategory, cf. [7] Section 2.1.

Example 3.3. Let **Cat** denote the *category of small categories*, where the objects are small categories, the hom-categories $\mathbf{Cat}(\mathscr{A},\mathscr{B})$ are functors between small categories $\mathscr{A}$ and $\mathscr{B}$, and the morphisms in $\mathbf{Cat}(\mathscr{A},\mathscr{B})$ are natural transformations between functors. This specifies **Cat** to be a 2-category.

3.2 Pseudofunctor, Pseudotransform, and Modification

Definition 3.4 (Pseudofunctor)**.** Given two 2-categories $\mathscr{A}, \mathscr{B}$, a *pseudofunctor* $F : \mathscr{A} \to \mathscr{B}$ equips with an 0-cell map, functors between hom-categories $F_{X,Y} : \mathscr{A}(X,Y) \to \mathscr{B}(F(X),F(Y))$, an invertible 2-cell $\phi_{X,Y,Z}$ between compositions

$$\begin{tikzcd}
\mathscr{A}(Y,Z)\times\mathscr{A}(X,Y) \arrow[r, "F_{Y,Z}\times F_{X,Y}"] \arrow[d, "\circ"'] & \mathscr{B}(F(Y),F(Z))\times\mathscr{B}(F(X),F(Y)) \arrow[d, "\circ"] \arrow[dl, Rightarrow, "\phi_{X,Y,Z}"'] \\
\mathscr{A}(X,Z) \arrow[r, "F_{X,Z}"'] & \mathscr{B}(F(X),F(Z))
\end{tikzcd}$$

and another invertible 2-cell ψ_X between identities

$$\begin{tikzcd}
F(X) \arrow[r, bend left, "\mathrm{id}_{F(X)}"] \arrow[r, bend right, "F_{X,X}(\mathrm{id}_X)"'] \arrow[r, phantom, "\Downarrow\psi_X"] & F(X)
\end{tikzcd}$$

that satisfies:

(i) **(Associativity)** The following two pasting diagrams are equal:

$$\begin{tikzcd}
\mathscr{A}(Z,W)\times\mathscr{A}(Y,Z)\times\mathscr{A}(X,Y) \arrow[r, "\prod F_{-,-}"] \arrow[d, "\circ\times\mathrm{id}"'] & \mathscr{B}(F(Z),F(W))\times\mathscr{B}(F(Y),F(Z))\times\mathscr{B}(F(X),F(Y)) \arrow[d, "\circ\times\mathrm{id}"] \arrow[dl, Rightarrow, "\phi_{Y,Z,W}\times\mathrm{id}"'] \\
\mathscr{A}(Y,W)\times\mathscr{A}(X,Y) \arrow[r, "F_{Y,W}\times F_{X,Y}"] \arrow[d, "\circ"'] & \mathscr{B}(F(Y),F(W))\times\mathscr{B}(F(X),F(Y)) \arrow[d, "\circ"] \arrow[dl, Rightarrow, "\phi_{X,Y,W}"'] \\
\mathscr{A}(X,W) \arrow[r, "F_{X,W}"'] & \mathscr{B}(F(X),F(W))
\end{tikzcd}$$

equals

$$\begin{tikzcd}
\mathscr{A}(Z,W)\times\mathscr{A}(Y,Z)\times\mathscr{A}(X,Y) \arrow[r, "\prod F_{-,-}"] \arrow[d, "\mathrm{id}\times\circ"'] & \mathscr{B}(F(Z),F(W))\times\mathscr{B}(F(Y),F(Z))\times\mathscr{B}(F(X),F(Y)) \arrow[d, "\mathrm{id}\times\circ"] \arrow[dl, Rightarrow, "\mathrm{id}\times\phi_{X,Y,Z}"'] \\
\mathscr{A}(Z,W)\times\mathscr{A}(X,Z) \arrow[r, "F_{Z,W}\times F_{X,Z}"] \arrow[d, "\circ"'] & \mathscr{B}(F(Z),F(W))\times\mathscr{B}(F(X),F(Z)) \arrow[d, "\circ"] \arrow[dl, Rightarrow, "\phi_{X,Z,W}"'] \\
\mathscr{A}(X,W) \arrow[r, "F_{X,W}"'] & \mathscr{B}(F(X),F(W))
\end{tikzcd}$$

(ii) **(Unitality)** The following two unit diagrams commute:

$$\begin{tikzcd}
\mathscr{A}(X,Y)\times\mathscr{A}(X,X) \arrow[r, "F_{X,Y}\times F_{X,X}"] & \mathscr{B}(F(X),F(Y))\times\mathscr{B}(F(X),F(X)) \arrow[dd, "\circ"] \arrow[ddl, Rightarrow, "\mathrm{id}*\psi_X"'] \\
\mathscr{A}(X,Y)\times * \arrow[u, "\mathrm{id}\times\mathrm{id}_X"] \arrow[d, "F_{X,Y}\times\mathrm{id}_{F(X)}"'] & \\
\mathscr{B}(F(X),F(Y))\times\mathscr{B}(F(X),F(X)) \arrow[r, "\circ"'] & \mathscr{B}(F(X),F(Y))
\end{tikzcd}$$

$$\begin{tikzcd}
\mathscr{A}(Y,Y)\times\mathscr{A}(X,Y) \arrow[r, "F_{Y,Y}\times F_{X,Y}"] & \mathscr{B}(F(Y),F(Y))\times\mathscr{B}(F(X),F(Y)) \arrow[dd, "\circ"] \arrow[ddl, Rightarrow, "\psi_Y*\mathrm{id}"'] \\
*\times\mathscr{A}(X,Y) \arrow[u, "\mathrm{id}_Y\times\mathrm{id}"] \arrow[d, "\mathrm{id}_{F(Y)}\times F_{X,Y}"'] & \\
\mathscr{B}(F(Y),F(Y))\times\mathscr{B}(F(X),F(Y)) \arrow[r, "\circ"'] & \mathscr{B}(F(X),F(Y))
\end{tikzcd}$$

where $\mathrm{id}_X : * \to \mathscr{A}(X,X)$ is the functor indicating identity of X.

If ψ_X is identity 2-cell for all 0-cell X, then the pseudofunctor F is called *normal*.

Definition 3.5 (Pseudotransform)**.** Fix 2-categories $\mathscr{A}, \mathscr{B}$, let $(F,\phi,\psi), (G,\varphi,\vartheta) : \mathscr{A} \to \mathscr{B}$ be two pseudofunctors. A *pseudotransform* $\mu : (F,\phi,\psi) \to (G,\varphi,\vartheta)$ collects 1-cells $\mu_X : F(X) \to G(X)$ for each 0-cell $X \in \mathscr{A}$, and for each 1-cell $f : X \to Y$ in $\mathscr{A}$, associates a 2-cell μf such that

$$\begin{tikzcd}
F(X) \arrow[r, "\mu_X"] \arrow[d, "Ff"'] & G(X) \arrow[d, "Gf"] \arrow[dl, Rightarrow, "\mu f"'] \\
F(Y) \arrow[r, "\mu_Y"'] & G(Y)
\end{tikzcd}$$

commutes. They also satisfy:

(i) (**Associativity**) Let $f : X \to Y$, $g : Y \to Z$ be 1-cells in $\mathscr{A}$, the following two pasting diagrams are equal:

$$
\begin{tikzcd}
F(X) \arrow[rr, "\mu_X"] \arrow[dr, "Ff"] \arrow[dd, "F(g\circ f)"'] & & G(X) \arrow[dr, "Gf"] \arrow[dl, Rightarrow, "\mu f"'] & \\
\arrow[r, phantom, "\overset{\phi_{g,f}}{\Longleftarrow}"] & F(Y) \arrow[rr, "\mu_Y"] \arrow[dl, "Fg"] & & G(Y) \arrow[dl, "Gg"] \arrow[dlll, Rightarrow, "\mu g"'] \\
F(Z) \arrow[rr, "\mu_Z"'] & & G(Z) &
\end{tikzcd}
\quad = \quad
\begin{tikzcd}
F(X) \arrow[r, "\mu_X"] \arrow[dd, "F(g\circ f)"'] & G(X) \arrow[dr, "Gf"] \arrow[dd, "G(g\circ f)"] \arrow[ddl, Rightarrow, "\mu(g\circ f)"'] & \\
& \arrow[r, phantom, "\overset{\varphi_{g,f}}{\Longleftarrow}"] & G(Y) \arrow[dl, "Gg"] \\
F(Z) \arrow[r, "\mu_Z"'] & G(Z) &
\end{tikzcd}
$$

(ii) (**Unitality**) The following pasting diagrams are equal:

$$
\begin{tikzcd}
F(X) \arrow[r, "\mu_X"] \arrow[d, bend right=50, "F(\mathrm{id}_X)"'] \arrow[d, bend left=50, "\mathrm{id}_{F(X)}"] \arrow[d, Rightarrow, "\psi_X"'] & G(X) \arrow[d, "\mathrm{id}_{G(X)}"] \\
F(X) \arrow[r, "\mu_X"'] & G(X)
\end{tikzcd}
\quad = \quad
\begin{tikzcd}
F(X) \arrow[r, "\mu_X"] \arrow[d, "F(\mathrm{id}_X)"'] & G(X) \arrow[dl, Rightarrow, "\mu(\mathrm{id}_X)"'] \arrow[d, bend right=50, "G(\mathrm{id}_X)"'] \arrow[d, bend left=50, "\mathrm{id}_{G(X)}"] \arrow[d, Rightarrow, "\vartheta_X"] \\
F(X) \arrow[r, "\mu_X"'] & G(X)
\end{tikzcd}
$$

Definition 3.6 (Modification)**.** Continue with the same notation as above. Let $\mu, \nu : (F, \phi, \psi) \to (G, \varphi, \vartheta)$ be two pseudotransforms, a *modification* $\lambda : \mu \Rightarrow \nu$ is a collection of 2-cells $\lambda_X : \mu_X \Rightarrow \nu_X$ for each 0-cell $X \in \mathscr{A}$, such that the following two pasting diagrams are equal, for each 1-cell $f : X \to Y$ in $\mathscr{A}$:

$$
\begin{tikzcd}
F(X) \arrow[r, bend left=50, "\mu_X"] \arrow[r, bend right=50, "\nu_X"'] \arrow[r, Rightarrow, "\lambda_X"] \arrow[d, "Ff"'] & G(X) \arrow[d, "Gf"] \arrow[dl, Rightarrow, "\nu f"] \\
F(Y) \arrow[r, "\nu_Y"'] & G(Y)
\end{tikzcd}
\quad = \quad
\begin{tikzcd}
F(X) \arrow[r, "\mu_X"] \arrow[d, "Ff"'] & G(X) \arrow[d, "Gf"] \arrow[dl, Rightarrow, "\mu f"'] \\
F(Y) \arrow[r, bend left=50, "\mu_Y"] \arrow[r, bend right=50, "\nu_Y"'] \arrow[r, Rightarrow, "\lambda_Y"] & G(Y)
\end{tikzcd}
$$

3.3 G-Categories

Definition 3.7. A small category $\mathscr{A}$ is a *G-category*, if the object and morphism sets $\mathrm{Ob}(\mathscr{A}), \mathrm{Mor}(\mathscr{A})$ are both (left) G-sets, such that the following four structure maps (S, T, I, C) (stands for *source, target, identity, and composition*) are all G-equivariant:

- $S, T : \mathrm{Mor}(\mathscr{A}) \to \mathrm{Ob}(\mathscr{A})$ by $S(X \xrightarrow{f} Y) := X$, and $T(X \xrightarrow{f} Y) := Y$.
- $I : \mathrm{Ob}(\mathscr{A}) \to \mathrm{Mor}(\mathscr{A})$ by $I(X) := \mathrm{id}_X$.
- $C : \mathrm{Mor}(\mathscr{A}) \times_{S,T} \mathrm{Mor}(\mathscr{A}) \to \mathrm{Mor}(\mathscr{A})$ by $C(g, f) := g \circ f$.

Here, $\mathrm{Mor}(\mathscr{A}) \times_{S,T} \mathrm{Mor}(\mathscr{A}) := \{(g, f) \in \mathrm{Mor}(\mathscr{A})^2 \mid S(g) = T(f)\}$, precisely the pair of morphisms that're composable, and it endows G-set structure via restricting the diagonal G-action on $\mathrm{Mor}(\mathscr{A})^2$. This is well-defined due to the assumption S, T are G-maps.

Another equivalent characterization can be viewed as a group homomorphism $G \to \mathrm{Aut}_{\mathbf{Cat}}(\mathscr{A})$, which assigns each $g \in G$ an automorphism functor on $\mathscr{A}$ such that composition respects G's multiplication structure, and identity gets sent to the identity functor. So, from now on we view each element $g \in G$ as a functor, whenever a category is specified as a G-category.

Definition 3.8. Let $\mathbf{Cat}_G$ denotes the 2-category of small G-categories, with (nonequivariant) functors as 1-cells, natural transformations as 2-cells. The hom-category between any two G-categories $\mathbf{Cat}_G(\mathscr{A}, \mathscr{B})$ also has a natural G-action via conjugation. Which, the G-fixed subcategory of the hom-category $[\mathbf{Cat}_G(\mathscr{A}, \mathscr{B})]^G$ precisely describes all G-functor, or the functors $F : \mathscr{A} \to \mathscr{B}$ such that $g \circ F = F \circ g$ for all $g \in G$.

Another important notion is the twisted G-action on products when given a group homomorphism $\alpha : G \to \Sigma_n$. Given a category $\mathscr{A}$, the n-fold product $\mathscr{A}^{\mathbf{n}}$ has a natural left Σ_n-action via permuting the coordinates; if $\mathscr{A}$ is a G-category, $\mathscr{A}^{\mathbf{n}}$ also endows a natural diagonal G-action. The two coordinates to give a natural left $(G \times \Sigma_n)$-action on $\mathscr{A}^{\mathbf{n}}$.

Definition 3.9. Let $\mathscr{A}^{\mathbf{n}^\alpha}$ be a G-category with the underlying category being $\mathscr{A}^{\mathbf{n}}$, with the G-action defined as follow:

$$g \cdot_\alpha (X_1, ..., X_n) := (g \cdot X_{\alpha(g)^{-1}(1)}, ..., g \cdot X_{\alpha(g)^{-1}(n)})$$

Another characterization can be understood as follow: Given the graph subgroup $\Gamma_\alpha := \{(g, \alpha(g)) \in G \times \Sigma_n\}_{g \in G}$, the projection $\pi_G : G \times \Sigma_n \twoheadrightarrow G$ restricts to an isomorphism $\Gamma_\alpha \xrightarrow{\sim} G$. The α-twisted action is induced by the pullback along this isomorphism.

3.4 Chaotic-Forgetful Adjunction

Given any small category $\mathscr{A} \in \mathbf{Cat}$, its object $\mathrm{Ob}(\mathscr{A})$ forms a set; moreover, any functor $F : \mathscr{A} \to \mathscr{B}$ also has an underlying map of objects $F : \mathrm{Ob}(\mathscr{A}) \to \mathrm{Ob}(\mathscr{B})$. Conversely, given a set, it has certain category structure:

Definition 3.10. Given a set X, define $\mathscr{E}X$ to be the *chaotic category* with underlying object set $\mathrm{Ob}(\mathscr{E}X) := X$, by encoding a unique isomorphism $x \xrightarrow{\sim} y$ between any element $x, y \in X$. The composition

$$(x \xrightarrow{\sim} y \xrightarrow{\sim} z) = (x \xrightarrow{\sim} z) \tag{3.11}$$

using the uniqueness.

Given a set map $f : X \to Y$, define the functor $\mathscr{E}f : \mathscr{E}X \to \mathscr{E}Y$ so that the underlying object map is f, and for each morphism $x \xrightarrow{\sim} y$ in $\mathscr{E}X$, $\mathscr{E}f(x \to y) := f(x) \to f(y)$, the unique morphism between $f(x), f(y)$ in $\mathscr{E}Y$.

Inspecting the definition, we see $\mathscr{E} : \mathbf{Set} \to \mathbf{Cat}$ is a well-defined functor, called the *chaotic functor*.

Notice if a functor $F : \mathscr{A} \to \mathscr{B}$ has a chaotic target $\mathscr{B}$, then all the morphisms $f : X \to Y$ in $\mathscr{A}$ must be sent to the unique morphism $F(X) \to F(Y)$ in $\mathscr{B}$, hence F is dictated by the object map $F : \mathrm{Ob}(\mathscr{A}) \to \mathrm{Ob}(\mathscr{B})$. Such uniqueness specifies an adjunction $(\mathrm{Ob} \dashv \mathscr{E})$.

The consequence is the following natural isomorphism of homset for any two sets X, Y:

$$\mathbf{Set}(X, Y) = \mathbf{Set}(\mathrm{Ob}(\mathscr{E}X), Y) \cong \mathbf{Cat}(\mathscr{E}X, \mathscr{E}Y) \tag{3.12}$$

Remark 3.13. Let $\mathbf{Set}_G$ denotes the category of (left) G-sets, with all maps between G-sets (not necessarily equivariant) being the morphisms, the above adjunction can be promoted to $\mathrm{Ob} : \mathbf{Cat}_G \to \mathbf{Set}_G$, and $\mathscr{E} : \mathbf{Set}_G \to \mathbf{Cat}_G$. Also, similar notions hold when transfer from left to right G-sets/categories.

In this exposition we only need $\mathscr{E}$ to be a right adjoint, there are richer details demonstrated in [3],

4 Operads, Operadic Algebra, and Operad $\mathscr{P}$

4.1 Symmetric Monoidal Categories

Definition 4.1. A category $\mathcal{V}$ is a *symmetric monoidal category*, if there is a bifunctor $\otimes : \mathcal{V} \times \mathcal{V} \to \mathcal{V}$ (called *monoidal product*), and an object $\kappa \in \mathcal{V}$ (called *unit/monoidal unit*), together with the following natural isomorphisms:

- The *associator* $a : (\_{1} \otimes \_{2}) \otimes \_{3} \xrightarrow{\sim} \_{1} \otimes (\_{2} \otimes \_{3})$.
- The *left unit* $\lambda : \kappa \otimes \_ \xrightarrow{\sim} \mathrm{Id}_{\mathcal{V}}$ and the *right unit* $\rho : \_ \otimes \kappa \xrightarrow{\sim} \mathrm{Id}_{\mathcal{V}}$.
- The *Symmetric Braiding* $B : \_{1} \otimes \_{2} \xrightarrow{\sim} \_{2} \otimes \_{1}$, with extra property $B_{Y,X} : Y \otimes X \to X \otimes Y$ equals to $B^{-1}_{X,Y} : X \otimes Y \to Y \otimes X$.

They also satisfy the following axioms:

- **(Triangle)**

$$\begin{tikzcd}
(X \otimes \kappa) \otimes Y \arrow[rr, "a_{X,\kappa,Y}"] \arrow[dr, "\rho_X \otimes \mathrm{id}_Y"'] & & X \otimes (\kappa \otimes Y) \arrow[dl, "\mathrm{id}_X \otimes \lambda_Y"] \\
& X \otimes Y &
\end{tikzcd}$$

- **(Pentagon)**

$$\begin{tikzcd}
((X \otimes Y) \otimes Z) \otimes W \arrow[rr, "a_{X \otimes Y,Z,W}"] \arrow[d, "a_{X,Y,Z} \otimes \mathrm{id}_W"'] & & (X \otimes Y) \otimes (Z \otimes W) \arrow[d, "a_{X,Y,Z \otimes W}"] \\
(X \otimes (Y \otimes Z)) \otimes W \arrow[dr, "a_{X,Y \otimes Z,W}"'] & & X \otimes (Y \otimes (Z \otimes W)) \\
& X \otimes ((Y \otimes Z) \otimes W) \arrow[ur, "\mathrm{id}_X \otimes a_{Y,Z,W}"'] &
\end{tikzcd}$$

- **(Hexagon)**

$$\begin{tikzcd}
(X \otimes Y) \otimes Z \arrow[r, "a_{X,Y,Z}"] \arrow[d, "B_{X,Y} \otimes \mathrm{Id}_Z"'] & X \otimes (Y \otimes Z) \arrow[r, "B_{X,Y \otimes Z}"] & (Y \otimes Z) \otimes X \arrow[d, "a_{Y,Z,X}"] \\
(Y \otimes X) \otimes Z \arrow[r, "a_{Y,X,Z}"'] & Y \otimes (X \otimes Z) \arrow[r, "\mathrm{id}_Y \otimes B_{X,Z}"'] & Y \otimes (Z \otimes X)
\end{tikzcd}$$

Based on *Maclane's Coherence Theorem*, the above two axioms guarantees all finite sequences of a, λ, ρ and their inverses agree, namely there is a unique isomorphism $P_1 \to P_2$, where P_i are (possibly) different orders of performing product on $X_1, ..., X_n \in \mathcal{V}$. For details, cf. [11].

The category of sets **Set**, small categories **Cat** in 3.3, and small G-categories $\mathbf{Cat}_G$ in 3.8 are central examples for this, where the cartesian product $\mathscr{A} \times \mathscr{B}$ serves as monoidal product, and the terminal set/category $*$ serves as units for both.

4.2 Operads, Operadic Algebras

Definition 4.2. Given symmetric monoidal category $\mathcal{V}$, a Σ *operad* $\mathcal{C}$ in $\mathcal{V}$ is a collection of objects $\{\mathcal{C}(j) \in \mathcal{V}\}_{j\in\mathbb{N}}$, each $\mathcal{C}(j)$ with a right Σ_j-action (a group homomorphism $\Sigma_j^{\mathrm{op}} \to \mathrm{Aut}_{\mathcal{V}}(\mathcal{C}(j))$), a *unit map* $\eta : 1 \to \mathcal{C}(1)$, together with composition maps

$$\gamma : \mathcal{C}(k) \otimes \mathcal{C}(j_1) \otimes ... \otimes \mathcal{C}(j_k) \to \mathcal{C}(j_1 + ... + j_k)$$

that satisfies the following axioms:

- *Unit:*

$$\begin{tikzcd} \mathcal{C}(j) \otimes \kappa^{\otimes j} \arrow[rr, "\sim"] \arrow[dr, "\mathrm{id}_{\mathcal{C}(j)}\otimes\eta^{\otimes j}"'] & & \mathcal{C}(j) \\ & \mathcal{C}(j) \otimes \mathcal{C}(1)^{\otimes j} \arrow[ur, "\gamma"'] & \end{tikzcd}, \qquad \begin{tikzcd} \kappa \otimes \mathcal{C}(j) \arrow[rr, "\sim"] \arrow[dr, "\eta\otimes\mathrm{id}_{\mathcal{C}(j)}"'] & & \mathcal{C}(j) \\ & \mathcal{C}(1) \otimes \mathcal{C}(j) \arrow[ur, "\gamma"'] & \end{tikzcd}$$

- *Associativity:* Fix natural numbers k, $j = \sum_{s=1}^k j_s$, define index $g_s := j_1 + ... + j_s$ for $1 \le s \le k$. For each $i_1, ..., i_{g_k} = i_j$, with sum $i = \sum_{r=1}^j i_r$, and $h_s := i_{g_{(s-1)}+1} + ... + i_{g_s}$ (so $i = \sum_{s=1}^k h_s$ also), the following diagram commutes:

$$\begin{tikzcd} \left[\mathcal{C}(k) \otimes \left(\bigotimes_{s=1}^k \mathcal{C}(j_s)\right)\right] \otimes \left(\bigotimes_{r=1}^j \mathcal{C}(i_r)\right) \arrow[r, "\gamma\otimes\mathrm{id}"] \arrow[dd, "\text{shuffle}"'] & \mathcal{C}(j) \otimes \left(\bigotimes_{r=1}^j \mathcal{C}(i_r)\right) \arrow[d, "\gamma"] \\ & \mathcal{C}(i) \\ \mathcal{C}(k) \otimes \left[\bigotimes_{s=1}^k \left(\mathcal{C}(j_s) \otimes \left(\bigotimes_{q=1}^{j_s} \mathcal{C}(i_{g_{(s-1)}+q})\right)\right)\right] \arrow[r, "\mathrm{id}_{\mathcal{C}(k)}\otimes(\bigotimes_s \gamma)"'] & \mathcal{C}(k) \otimes \left(\bigotimes_{s=1}^k \mathcal{C}(h_s)\right) \arrow[u, "\gamma"'] \end{tikzcd}$$

- *Equivariance:*

  - Given $\sigma \in \Sigma_k$, $j_1 + ... + j_k = \Sigma_j$, let $\sigma(j_1, ..., j_k) \in \Sigma_j$ denotes "permutation of k-blocks of letters", by $(j_1, ..., j_k) \mapsto (j_{\sigma^{-1}(1)}, ..., j_{\sigma^{-1}(k)})$. Then, the following diagram commutes:

$$\begin{tikzcd} \mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^k \mathcal{C}(j_r)\right) \arrow[r, "\sigma\otimes\sigma^{-1}"] \arrow[d, "\gamma"'] & \mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^k \mathcal{C}(j_{\sigma(r)})\right) \arrow[d, "\gamma"] \\ \mathcal{C}(j) \arrow[r, "\sigma(j_{\sigma(1)},...,j_{\sigma(k)})"'] & \mathcal{C}(j) \end{tikzcd}$$

  - Given $\tau_r \in \Sigma_{j_r}$, denote $\tau_1 \oplus ... \oplus \tau_k \in \Sigma_j$ by letting each τ_r permuting the j_rth block in j. Then, the following diagram commutes:

$$\begin{tikzcd} \mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^k \mathcal{C}(j_r)\right) \arrow[r, "\mathrm{id}_{\mathcal{C}(k)}\otimes(\bigotimes_r \tau_r)"] \arrow[d, "\gamma"'] & \mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^k \mathcal{C}(j_r)\right) \arrow[d, "\gamma"] \\ \mathcal{C}(j) \arrow[r, "\tau_1\oplus...\oplus\tau_k"'] & \mathcal{C}(j) \end{tikzcd}$$

If $\mathcal{C}(0) = \kappa$ (with $\gamma : \mathcal{C}(0) \to \mathcal{C}$ being $\mathrm{id}_{\mathcal{C}(0)}$), the operad is *reduced.*

Definition 4.3. Given an operad $\mathscr{C}$ over a symmetric monoidal category $\mathcal{V}$, a $\mathscr{C}$*-algebra* is an object $A \in \mathcal{V}$, with collections of morphisms $\theta : \mathscr{C}(j) \otimes A^j \to A$, that satisfies the following axioms:

- *Associativity:* Given $j = \sum_{i=1}^k j_i$, the following diagram commutes:

$$\begin{tikzcd}
\left[\mathscr{C}(k) \otimes \left(\bigotimes_{i=1}^k \mathscr{C}(j_i)\right)\right] \otimes A^{\otimes j} \arrow[r, "\gamma\otimes\mathrm{id}_{A^{\otimes j}}"] \arrow[dd, "\mathrm{shuffle}"'] & \mathscr{C}(j) \otimes A^{\otimes j} \arrow[d, "\theta"] \\
 & A \\
\mathscr{C}(k) \otimes \left[\bigotimes_{i=1}^k \left(\mathscr{C}(j_i) \otimes A^{\otimes j_i}\right)\right] \arrow[r, "\mathrm{id}_{\mathscr{C}(k)}\otimes\theta^k"'] & \mathscr{C}(k) \otimes A^{\otimes k} \arrow[u, "\theta"']
\end{tikzcd}$$

  (Note: This part uses the isomorphism $A^{\otimes j} \cong \bigotimes_{i=1}^k A^{\otimes j_i}$).

- *Unit:* The following diagram commutes:

$$\begin{tikzcd}
\kappa \otimes A \arrow[rr, "\sim"] \arrow[dr, "\eta\otimes\mathrm{id}_A"'] & & A \\
 & \mathscr{C}(1) \otimes A \arrow[ur, "\theta"'] &
\end{tikzcd}$$

- *Eauivariance:* Given any permutation $\sigma \in \Sigma_j$, the following diagram commutes:

$$\begin{tikzcd}
\mathscr{C}(j) \otimes A^{\otimes j} \arrow[rr, "\sigma\otimes\sigma^{-1}"] \arrow[dr, "\theta"'] & & \mathscr{C}(j) \otimes A^{\otimes j} \arrow[dl, "\theta"] \\
 & A &
\end{tikzcd}$$

Example 4.4. The *associativity operad* $\mathscr{M}$ in **Set** is defined as $\mathscr{M}(j) := \Sigma_j$ (the symmetric group for cardinality j), which endows a natural right Σ_j-action via right multiplication. The operadic structure map is given by

$$\gamma : \Sigma_k \times \prod_{r=1}^k \Sigma_{j_r} \to \Sigma_j, \quad \gamma(\sigma; \tau_1, ..., \tau_k) := \tau_{\sigma^{-1}(1)} \oplus ... \oplus \tau_{\sigma^{-1}(k)} \circ \sigma(j_{\sigma(1)}, ..., j_{\sigma(k)}) \tag{4.5}$$

where $j = \sum_r j_r$, $\sigma(j_1, ..., j_k) \in \Sigma_j$ denotes the block permutation of $j_1, ..., j_k$ within j (in order), and $\tau_1 \oplus ...\tau_k$ is the image of the embedding $\oplus : \prod_{r=1}^k \Sigma_{j_r} \to \Sigma_j$, by viewing the rth block as finite set of cardinality j_r. Inspecting the definitions, the algebra over $\mathscr{M}$ precisely describes the set theoretic monoids. For more details, cf. [3] Section 4.1.

4.3 Permutativity Operad $\mathscr{P}$

Now, recall that the chaotic functor $\mathscr{E} : \mathbf{Set} \to \mathbf{Cat}$ is a right adjoint of $\mathrm{Ob} : \mathbf{Cat} \to \mathbf{Set}$, in particular it preserves products. Hence, the chaotic categorification $\mathscr{P}(j) := \mathscr{E}\mathscr{M}(j)$ also has a natural operadic structure, via the identification

$$\mathscr{E}\mathscr{M}(k) \times \prod_{r=1}^k \mathscr{E}\mathscr{M}(j_r) \cong \mathscr{E}\left(\mathscr{M}(k) \times \prod_{r=1}^k \mathscr{M}(j_r)\right) \xrightarrow{\mathscr{E}\gamma} \mathscr{E}\mathscr{M}(j). \tag{4.6}$$

This is called the *permutativity operad in* **Cat**, or the *Barratt-Eccles operad.* As an abuse of notation, the operadic structure map of $\mathscr{P}$ is also denoted γ.

5 G-Operad $\mathscr{P}_G$ and its Pseudoalgebra

5.1 The G-Operad $\mathscr{P}_G$

If view each $\mathscr{P}(j)$ as a G-trivial G-category, it's also an operad in $\mathbf{Cat}_G$. Its underlying algebra is called the *naive permutative G-category*, as specified in [3] Section 4.1, it produces a naive G-spectrum under equivariant infinite loop space theory. In the same source Section 4.2, a richer equivariant structure is introduced which produces the *equivariant Barratt Eccles G-operad $\mathscr{P}_G$* as follow:

Definition 5.1. The jth category $\mathscr{P}_G(j) := \mathbf{Cat}(\mathscr{E}G, \mathscr{P}(j))$, with the operadic structure map determined by the fact that $\mathbf{Cat}(\mathscr{E}G, \_)$ preserves product:

$$\mathbf{Cat}(\mathscr{E}G, \mathscr{P}(k)) \times \prod_{r=1}^{k} \mathbf{Cat}(\mathscr{E}G, \mathscr{P}(j_r)) \cong \mathbf{Cat}\left(\mathscr{E}G, \mathscr{P}(k) \times \prod_{r=1}^{k} \mathscr{P}(j_r)\right) \xrightarrow{\gamma\circ\_} \mathbf{Cat}(\mathscr{E}, \mathscr{P}(j)) \tag{5.2}$$

Abuse of notation, we'll denote the operadic structure map as γ again.

Another formulation can be done by defining $P_G(j) := \mathbf{Set}(G, \Sigma_j)$ in $\mathbf{Set}$, then utilize chaotic functor promoting this to a chaotic operad $\mathscr{E}\mathbf{Set}(G, \Sigma) \cong \mathscr{P}_G$.

5.2 The 2-Category $\mathscr{P}_G$-PsAlg

The general notion of operadic pseudoalgebra is defined in [2], [5], and [8]. Here we only take the case when the operad is $\mathscr{P}_G$, together with some description of G-equivariance.

Definition 5.3 ($\mathscr{P}_G$-pseudoalgebra)**.** A $\mathscr{P}_G$-pseudoalgebra is a G-category $\mathscr{A}$ and operad action functors

$$\theta_j : \mathscr{P}_G(j) \times \mathscr{A}^j \to \mathscr{A}$$

for each $n \in \mathbb{N}$, together with an invertible natural transformation ϕ_1 in the diagram

$$\begin{tikzcd} \mathscr{A} \arrow[r, "\iota"] \arrow[dr, "\mathrm{Id}_{\mathscr{A}}"'] \arrow[dr, phantom, shift left=3, "\Leftarrow \phi_1"] & \mathscr{P}_G(1) \times \mathscr{A} \arrow[d, "\theta_1"] \\ & C \end{tikzcd} \tag{5.4}$$

An invertible natural transformation $\phi = \phi(k; j_1, ..., j_k)$ for each diagram

$$\begin{tikzcd} \mathscr{P}_G(k) \times \left(\prod_{r=1}^{k} \mathscr{P}_G(j_r) \times \mathscr{A}^{j_r}\right) \arrow[rr, "\mathrm{id}\times\prod_r \theta_{j_r}"] \arrow[dd, "\text{shuffle}"'] & & \mathscr{P}_G(k) \times \mathscr{A}^k \arrow[d, "\theta_k"] \arrow[ddll, Rightarrow, "\phi"] \\ & & \mathscr{A} \\ \mathscr{P}_G(k) \times \prod_{r=1}^{k} \mathscr{P}_G(j_r) \times \mathscr{A}^{j_+} \arrow[rr, "\gamma\times\mathrm{id}"'] & & \mathscr{P}_G(j_+) \times \mathscr{A}^{j_+} \arrow[u, "\theta_{j_+}"'] \end{tikzcd} \tag{5.5}$$

such that the following strict equivariance hold for θ_j and ϕ:

(i) **(Equivariance of θ_j)** The following two diagrams commutes for $\sigma \in \Sigma_j$ and $g \in G$:

$$\begin{tikzcd} \mathscr{P}_G(j) \times \mathscr{A}^j \arrow[rr, "\sigma\times\sigma^{-1}"] \arrow[dr, "\theta_j"'] & & \mathscr{P}_G(j) \times \mathscr{A}^j \arrow[dl, "\theta_j"] \\ & \mathscr{A} & \end{tikzcd}, \qquad \begin{tikzcd} \mathscr{P}_G(j) \times \mathscr{A}^j \arrow[r, "g\times g^{-1}"] \arrow[d, "\theta_j"'] & \mathscr{P}_G(j) \times \mathscr{A}^j \arrow[d, "\theta_j"] \\ \mathscr{A} \arrow[r, "g^{-1}"'] & \mathscr{A} \end{tikzcd} \tag{5.6}$$

The left is the operadic equivariance, the right is G-equivariance by asserting a product G-action on $\mathscr{P}(j) \times \mathscr{A}^j$.

(ii) **(Equivariance of ϕ)** The following equalities hold when given $\sigma \in \Sigma_k$, and $\tau_r \in \Sigma_{j_r}$:

$$\phi(k; j_1, ..., j_k) = \phi(k; j_{\sigma(1)}, ..., j_{\sigma_k}) \circ (\sigma \times \sigma^{-1})$$

$$\phi(k; j_1, ..., j_k) = \phi(k; j_1, ..., j_k) \circ (\mathrm{id} \times \prod_r (\tau_r \times \tau_r^{-1}))$$

They also need to satisfy the following coherence properties:

(iii) **(Operadic Composition)** The following pasting diagrams are equal, given $j_+ = \sum j_r$, $i_r = \sum_s i_{r,s}$, and $i_+ = \sum_{r,s} i_{r,s}$ with $1 \le r \le k$ and $1 \le s \le j_r$:

$$\begin{tikzcd}
\mathscr{P}_G(k) \times \prod_r \left(\mathscr{P}_G(j_r) \times \prod_s \left(\mathscr{P}_G(i_{r,s}) \times \mathscr{A}^{i_{r,s}}\right)\right) \arrow[r, "\mathrm{id}\times\prod(\mathrm{id}\times\theta_{i_{r,s}})"] \arrow[dd, "\gamma\times\mathrm{id}"'] & Q_G(k) \times \prod_r \left(\mathscr{P}_G(j_r) \times \mathscr{A}^{j_r}\right) \arrow[dd, "\gamma"'] \arrow[dr, "\mathrm{id}\times\prod_{\theta_{j_r}}"] & \\
& & \mathscr{P}_G(k) \times \mathscr{A}^k \arrow[dl, Rightarrow, "\phi"] \arrow[dd, "\theta_k"] \\
\mathscr{P}_G(j_+) \times \prod_{r,s} \left(\mathscr{P}_G(i_{r,s}) \times \mathscr{A}^{i_{r,s}}\right) \arrow[r, "\mathrm{id}\times\prod\theta_{i_{r,s}}"] \arrow[dr, "\gamma\times\mathrm{id}"'] & \mathscr{P}_G(j_+) \times \mathscr{A}^{j_+} \arrow[d, Rightarrow, "\phi"] \arrow[dr, "\theta_{j_+}"] & \\
& \mathscr{P}_G(i_+) \times \mathscr{A}^{i_+} \arrow[r, "\theta_{j_+}"'] & \mathscr{A}
\end{tikzcd}$$

equals

$$\begin{tikzcd}
\mathscr{P}_G(k) \times \prod_r \left(\mathscr{P}_G(j_r) \times \prod_s \left(\mathscr{P}_G(i_{r,s}) \times \mathscr{A}^{i_{r,s}}\right)\right) \arrow[r, "\mathrm{id}\times\prod(\mathrm{id}\times\prod\theta_{i_{r,s}})"] \arrow[dd, "\gamma\times\mathrm{id}"'] \arrow[dr, "\mathrm{id}\times\prod(\gamma\times\mathrm{id})"'] & \mathscr{P}_G(k) \times \prod_r \left(\mathscr{P}_G(j_r) \times \mathscr{A}^{j_r}\right) \arrow[d, Rightarrow, "\mathrm{id}\times\prod\phi"] \arrow[dr, "\mathrm{id}\times\prod\theta_{j_r}"] & \\
& \mathscr{P}_G(k) \times \prod_r \left(\mathscr{P}_G(i_r) \times \mathscr{A}^{i_r}\right) \arrow[r, "\mathrm{id}\times\prod\gamma"'] \arrow[dd, "\gamma\times\mathrm{id}"'] & \mathscr{P}_G(k) \times \mathscr{A}^k \arrow[ddl, Rightarrow, "\phi"] \arrow[dd, "\theta_k"] \\
\mathscr{P}_G(j_+) \times \prod_{r,s} \left(\mathscr{P}_G(i_{r,s}) \times \mathscr{A}^{i_{r,s}}\right) \arrow[dr, "\gamma\times\mathrm{id}"'] & & \\
& \mathscr{P}_G(i_+) \times \mathscr{A}^{i_+} \arrow[r, "\theta_{i_+}"'] & \mathscr{A}
\end{tikzcd}$$

The commuting square above is due to associativity of operad.

(iv) **(Operadic Identity)** The following two pairs of two pasting diagrams are equal:

$$
\begin{tikzcd}
\mathscr{P}_G(k)\times\mathscr{A}^k \arrow[r, "\mathrm{id}\times\iota^k"] & \mathscr{P}_G(k)\times(\mathscr{P}_G(1)\times\mathscr{A})^k \arrow[r, "\mathrm{id}\times\theta_1^k"] \arrow[d, "\mathrm{shuffle}"'] & \mathscr{P}_G(k)\times\mathscr{A}^k \arrow[dd, "\theta_k"] \arrow[ddl, Rightarrow, "\phi"] \\
& \mathscr{P}_G(k)\times\mathscr{P}_G(1)^k\times\mathscr{A}^k \arrow[d, "\gamma\times\mathrm{id}"'] & \\
& \mathscr{P}_G(k)\times\mathscr{A}^k \arrow[r, "\theta_k"'] & \mathscr{A}
\end{tikzcd}
$$

equals

$$
\begin{tikzcd}
& \mathscr{P}_G(k)\times(\mathscr{P}_G(1)\times\mathscr{A})^k \arrow[dr, "\mathrm{id}\times\theta_1^k"] \arrow[d, Rightarrow, "\mathrm{id}\times\prod\phi"] & & \\
\mathscr{P}_G(k)\times\mathscr{A}^k \arrow[ur, "\mathrm{id}\times\iota^k"] \arrow[rr, "\mathrm{id}"'] \arrow[d, "\mathrm{id}\times\iota^k"'] \arrow[dr, "\iota^k"] & {} & \mathscr{P}_G(k)\times\mathscr{A}^k \arrow[r, "\theta_k"] & \mathscr{A} \\
\mathscr{P}_G(k)\times(\mathscr{P}_G(1)\times\mathscr{A})^k \arrow[r, "\mathrm{shuffle}"'] & \mathscr{P}_G(k)\times\mathscr{P}_G(1)^k\times\mathscr{A}^k \arrow[ur, "\gamma\times\mathrm{id}"'] & &
\end{tikzcd}
$$

(The two triangles right above commutes due to the unit axioms of operad).

Also:

$$
\begin{tikzcd}
\mathscr{P}_G(k)\times\mathscr{A}^k \arrow[r, "\iota"] \arrow[dr, "\mathrm{id}"'] & \mathscr{P}_G(1)\times\mathscr{P}_G(k)\times\mathscr{A}^k \arrow[r, "\mathrm{id}\times\theta_k"] \arrow[d, "\gamma\times\mathrm{id}"'] & \mathscr{P}_G(1)\times\mathscr{A} \arrow[d, "\theta_1"] \arrow[dl, Rightarrow, "\phi"] \\
& \mathscr{P}_G(k)\times\mathscr{A}^k \arrow[r, "\theta_k"'] & \mathscr{A}
\end{tikzcd}
$$

equals

$$
\begin{tikzcd}
\mathscr{P}_G(k)\times\mathscr{A}^k \arrow[r, "\iota"] \arrow[d, "\theta_k"'] & \mathscr{P}_G(1)\times\mathscr{P}_G(k)\times\mathscr{A}^k \arrow[d, "\mathrm{id}\times\theta_k"] \\
\mathscr{A} \arrow[r, "\iota"] \arrow[dr, "\mathrm{id}"'] & \mathscr{P}_G(1)\times\mathscr{A} \arrow[d, "\theta_1"] \arrow[l, Rightarrow, shift left=2, "\phi_1"] \\
& \mathscr{A}
\end{tikzcd}
$$

The first diagram's triangle commutes due to he operadic unit axiom.

Remark 5.7. In [2] Corner and Gurski includes unital natural isomorphisms, while in [5] it requires strict unitality. The main result in nonequivariant case of [8] alleviates the result from strict unitality.x

Definition 5.8 ($\mathscr{P}_G$-pseudomorphism)**.** A $\mathscr{P}_G$-pseudomorphism $(\mathbb{F}, \zeta) : (\mathscr{A}, \theta, \phi) \to (\mathscr{B}, \vartheta, \varphi)$ consists of functor $\mathbb{F} : \mathscr{A} \to \mathscr{B}$, such that for each j in the follogin diagram commutes up to an invertible natural transformation ζ_j:

$$\begin{tikzcd}
\mathscr{P}_G(j) \times \mathscr{A}^j \arrow[r, "\mathrm{id}\times\mathbb{F}^j"] \arrow[d, "\theta_j"'] & \mathscr{P}_G(j) \times \mathscr{B}^j \arrow[d, "\vartheta_j"] \arrow[dl, phantom, "\Swarrow \zeta_j"] \\
\mathscr{A} \arrow[r, "\mathbb{F}"'] & \mathscr{B}
\end{tikzcd}$$

Moreover, they satisfy the following properties:

(i) **(Equivariance of ζ)** For each j,$\zeta_j = \zeta_j \circ (\sigma \times \sigma^{-1})$ for all $\sigma \in \Sigma_j$. **Check G-equivariance**

(ii) **(Preserves Identity)** The following two pasting diagrams are equal:

$$\begin{tikzcd}
\mathscr{A} \arrow[r, "\mathbb{F}"] \arrow[d, "\iota"'] & \mathscr{B} \arrow[d, "\iota"'] \arrow[ddr, bend left, "\mathrm{id}"] & \\
\mathscr{P}_G(1) \times \mathscr{A} \arrow[r, "\mathrm{id}\times\mathbb{F}"] \arrow[dr, "\theta_1"'] & \mathscr{P}_G(1) \times \mathscr{B} \arrow[dr, "\vartheta_1"] \arrow[r, phantom, "\overset{\varphi_1}{\Rightarrow}"] \arrow[d, phantom, "\Uparrow \zeta_1^{-1}"] & {} \\
& \mathscr{A} \arrow[r, "\mathbb{F}"'] & \mathscr{B}
\end{tikzcd}
=
\begin{tikzcd}
\mathscr{A} \arrow[r, "\mathbb{F}"] \arrow[d, "\iota"'] \arrow[dr, "\mathrm{id}"] & \mathscr{B} \arrow[dr, "\mathrm{id}"] & \\
\mathscr{P}_G(1) \times \mathscr{A} \arrow[r, "\theta_1"'] \arrow[ur, phantom, "\overset{\phi_1}{\Rightarrow}"] & \mathscr{A} \arrow[r, "\mathbb{F}"'] & \mathscr{B}
\end{tikzcd}$$

(iii) **(Preserves Composition)** The following two pasting diagrams are equal:

$$\begin{tikzcd}
\mathscr{P}_G(k) \times \prod_r \left(\mathscr{P}_G(j_r) \times \mathscr{A}^{j_r}\right) \arrow[rr, "\mathrm{id}\times\prod(\mathrm{id}\times\mathbb{F}^{j_r})"] \arrow[dd, "\gamma"'] \arrow[dr, "\mathrm{id}\times\prod\theta_{j_r}"] & & \mathscr{P}_G(k) \times \prod_r \left(\mathscr{P}_G(j_r) \times \mathscr{B}^{j_r}\right) \arrow[dr, "\mathrm{id}\times\prod\vartheta_{j_r}"] \arrow[dl, phantom, "\Swarrow \mathrm{id}\times\prod\zeta_{j_r}"] & \\
& \mathscr{P}_G(k) \times \mathscr{A}^k \arrow[rr, "\mathrm{id}\times\mathbb{F}^k"'] \arrow[dd, "\theta_k"] \arrow[dl, phantom, "\Swarrow \phi"] & & \mathscr{P}_G(k) \times \mathscr{B}^k \arrow[dd, "\vartheta_k"] \arrow[ddll, phantom, "\Swarrow \zeta_k"] \\
\mathscr{P}_G(j) \times \mathscr{A}^j \arrow[dr, "\theta_j"'] & & & \\
& \mathscr{A} \arrow[rr, "\mathbb{F}"'] & & \mathscr{B}
\end{tikzcd}$$

equals

$$\begin{tikzcd}
\mathscr{P}_G(k) \times \prod_r \left(\mathscr{P}_G(j_r) \times \mathscr{A}^{j_r}\right) \arrow[rr, "\mathrm{id}\times\prod(\mathrm{id}\times\mathbb{F}^{j_r})"] \arrow[dd, "\gamma"'] & & \mathscr{P}_G(k) \times \prod \left(\mathscr{P}_G(j_r) \times \mathscr{B}^{j_r}\right) \arrow[dd, "\gamma"'] \arrow[dr, "\mathrm{id}\times\prod\theta_{j_r}"] & \\
& & & \mathscr{P}_G(k) \times \mathscr{B}^k \arrow[dd, "\vartheta_k"] \arrow[dl, phantom, "\Swarrow \varphi"] \\
\mathscr{P}_G(j) \times \mathscr{A}^j \arrow[rr, "\mathrm{id}\times\mathbb{F}^j"] \arrow[dr, "\theta_j"'] & & \mathscr{P}_G(j) \times \mathscr{B}^j \arrow[dr, "\vartheta_j"] \arrow[dl, phantom, "\Swarrow \zeta_j"] & \\
& \mathscr{A} \arrow[rr, "\mathbb{F}"'] & & \mathscr{B}
\end{tikzcd}$$

Definition 5.9 ($\mathscr{P}_G$-2-cells)**.** Let $(\mathbb{F},\zeta),(\mathbb{G},\xi) : (\mathscr{A},\theta,\phi) \to (\mathscr{B},\vartheta,\varphi)$ be two $\mathscr{P}_G$-pseudomorphisms, a $\mathscr{P}_G$-2-cell $\lambda : (\mathbb{F},\zeta) \Rightarrow (\mathbb{G},\xi)$ is a natural transformation $\lambda : \mathbb{F} \to \mathbb{G}$, such that the following two pasting diagrams are equal:

$$
\begin{tikzcd}
\mathscr{P}_G(j) \times \mathscr{A}^j \arrow[r, bend left=40, "\mathrm{id}\times\mathbb{F}^j"] \arrow[r, Rightarrow, "\mathrm{id}\times\lambda^j"] \arrow[r, bend right=40, "\mathrm{id}\times\mathbb{G}^j"'] \arrow[d, "\theta_j"'] & \mathscr{P}_G(j) \times \mathscr{B}^j \arrow[d, "\vartheta_j"] \arrow[dl, Rightarrow, "\xi_j"] \\
\mathscr{A} \arrow[r, "\mathbb{G}"'] & \mathscr{B}
\end{tikzcd}
\quad = \quad
\begin{tikzcd}
\mathscr{P}_G(j) \times \mathscr{A}^j \arrow[r, "\mathrm{id}\times\mathbb{F}^j"] \arrow[d, "\theta_j"'] & \mathscr{P}_G(j) \times \mathscr{B}^j \arrow[d, "\vartheta_j"] \arrow[dl, Rightarrow, "\zeta_j"'] \\
\mathscr{A} \arrow[r, bend left=40, "\mathbb{F}"] \arrow[r, Rightarrow, "\lambda"'] \arrow[r, bend right=40, "\mathbb{G}"'] & \mathscr{B}
\end{tikzcd}
$$

6 Category $\mathscr{F}_G$ and its Pseudoalgebras

Within the equivariant infinite loop space machine, the purpose for $\mathscr{F}_G$-pseudoalgebras (after strictifying to algebras) are to construct $\mathscr{F}_G$-spaces, which are collections of G-spaces indexed over finite based G-sets (and morphisms between them encoding extra structure). The $\mathscr{F}_G$-spaces are primary inputs in producing a genuine G-spectra, and the other main character we wish to study the correspondance. For details of $\mathscr{F}_G$-spaces, cf. [14] Section 1 (which is named after Γ_G instead of $\mathscr{F}_G$).

The following definitions are equivariant generalizations of $\mathscr{F}$-pseudofunctors in [8], and we'll utilize similar notations.

Definition 6.1. $\mathscr{F}_G$ denotes the *category of based finite sets*, where the objects are $\mathbf{n}^\alpha$, set wise being the finite set $\{0, 1, ..., n\}$ with 0 fixed as the basepoint, and $\alpha : G \to \Sigma_n$ is a based G-action on the set.

The homset $\mathscr{F}_G(\mathbf{n}^\alpha, \mathbf{m}^\beta)$ consists of all the based set map, which the homset also endows with a G-action via conjugation. From now on, we'll view $\mathscr{F}_G$ as a 2-category, where each hom-category $\mathscr{F}_G(\mathbf{n}^\alpha, \mathbf{m}^\beta)$ is discrete with only identity 2-cells.

Furthermore, we define the following:

- $\Pi_G \subset \mathscr{F}_G$ denotes the subcategory where each morphism $\phi : \mathbf{n}^\alpha \to \mathbf{m}^\beta$ satisfies $|\phi^{-1}(j)| = 0, 1$ for all $j \neq 0$.
- $\Lambda_G, \Lambda_G^\perp, \Sigma_G \subset \Pi_G$ denotes the subcategory where each morphism is injective, surjective, and bijective respectively (in particular, $\Lambda_G \cap \Lambda_G^\perp = \Sigma_G$).
- $\mathscr{E}_G \subset \mathscr{F}_G$ denotes the subcategory where each morphism satisfies $\phi^{-1}(0) = \{0\}$.

We also denote the map $\phi_{\mathbf{n}^\alpha} : \mathbf{n}^\alpha \to \mathbf{1}$ (with $j \mapsto 1$ for all $j \neq 0$) in $\mathscr{E}$ as the *canonical product.*

6.1 2-Category $\mathscr{F}_G$-PsAlg

Definition 6.2 ($\mathscr{F}_G$-pseudoalgebra)**.** An $\mathscr{F}_G$-pseudoalgebra, is a pseudofunctor $\mathscr{B} : \mathscr{F}_G \to \mathbf{Cat}_G$ (cf. Definition 3.4), such that restricting to the subcategory $\Pi_G \subset \mathscr{F}_G$, $\mathscr{B}$ is a strict functor. Moreover, we require it to be reduced, namely $\mathscr{B}(0) = *$ (the terminal G-category), and each hom-functor

$$\mathscr{B}_{\mathbf{n}^\alpha, \mathbf{m}^\beta} : \mathscr{F}_G(\mathbf{n}^\alpha, \mathbf{m}^\beta) \to \mathbf{Cat}_G(\mathscr{B}(\mathbf{n}^\alpha), \mathscr{B}(\mathbf{m}^\beta))$$

is a G-functor.

Let $\delta_i : \mathbf{n}^\alpha \to \mathbf{1}$ be the ith projection (namely $i \mapsto 1$, and any $i \neq j \mapsto 0$), since it belongs to Π_G, it generates a functor $\mathscr{B}\delta_i : \mathscr{B}(\mathbf{n}^\alpha) \to \mathscr{B}(\mathbf{1})$. Hence, by universality of product, the collection $\mathscr{B}\delta_i$ $(1 \leq i \leq n)$ induces a functor $\delta : \mathscr{B}(\mathbf{n}^\alpha) \to \mathscr{B}(\mathbf{1})^{\mathbf{n}^\alpha}$ (where $\mathscr{B}(\mathbf{1})^{\mathbf{n}} \cong \mathscr{B}(\mathbf{1})^{\mathbf{n}^\alpha}$ nonequivariantly).

Definition 6.3 ($\mathscr{F}_{G,\mathrm{vs}}$ and $\mathscr{F}_{G,\mathbb{R}}$-pseudoalgebra)**.** An $\mathscr{F}_G$-pseudoalgebra $\mathscr{B}$ is *very special* (or an $\mathscr{F}_{G,\mathrm{vs}}$-pseudoalgebra), if $\delta : \mathscr{B}(\mathbf{n}^\alpha) \to \mathscr{B}(\mathbf{1})^{\mathbf{n}^\alpha}$ is an isomorphism for all $\mathbf{n}^\alpha \in \mathscr{F}_G$, and $\mathscr{B}$ is *strictly special* (or an $\mathscr{F}_{G,\mathbb{R}}$-pseudoalgebra), if all δ is in fact identity.

Definition 6.4 ($\mathscr{F}_G$-1-cell)**.** Given $\mathscr{A}, \mathscr{B}$ two $\mathscr{F}_G$-pseudoalgebras, a morphism (or $\mathscr{F}_G$-1-cell) between them is a pseudotransform $(\mathbb{F}, \zeta) : \mathscr{A} \to \mathscr{B}$ (cf. Definition 3.5), such that restricting to Π_G, it becomes an actual natural transformation. Moreover, it respects G-equivariance for every $g \in G$ and morphism $f \in \mathrm{Mor}(\mathscr{F}_G)$:

$$\zeta(g \circ f \circ g^{-1}) = g \circ \zeta(f) \circ g^{-1}$$

Definition 6.5 ($\mathscr{F}_G$-2-cell)**.** Given $\tilde{A}, \tilde{B}$ two $\mathscr{F}_G$-pseudoalgebras, and two $\mathscr{F}_G$-1-cells $(\mathbb{F}, \zeta), (\mathbb{G}, \xi) : \tilde{A} \to \tilde{\mathscr{B}}$, an $\mathscr{F}_G$-2-cell is a modification $\lambda : \mathbb{F} \Rightarrow \mathbb{G}$ (cf. Definition 3.6), such that for any morphism $f : \mathbf{n}^\alpha \to \mathbf{m}^\beta$ in $\mathscr{F}_G$, the following two pasting diagrams are equal:

$$\begin{tikzcd}
\mathscr{A}(\mathbf{n}^\alpha) \arrow[r, bend left, "\mathbb{F}_{\mathbf{n}^\alpha}"] \arrow[r, bend right, "\mathbb{G}_{\mathbf{n}^\alpha}"'] \arrow[r, phantom, "\overset{\lambda}{\Longrightarrow}"] \arrow[d, "\mathscr{A}f"'] & \mathscr{B}(\mathbf{n}^\alpha) \arrow[d, "\mathscr{B}f"] \arrow[dl, Rightarrow, "\xi f"] \\
\mathscr{A}(\mathbf{m}^\beta) \arrow[r, "\mathbb{G}_{\mathbf{m}^\beta}"'] & \mathscr{B}(\mathbf{m}^\beta)
\end{tikzcd}
\quad = \quad
\begin{tikzcd}
\mathscr{A}(\mathbf{n}^\alpha) \arrow[r, "\mathbb{F}_{\mathbf{n}^\alpha}"] \arrow[d, "\mathscr{B}f"'] & \mathscr{B}(\mathbf{n}^\alpha) \arrow[d, "\mathscr{B}f"] \arrow[dl, Rightarrow, "\zeta f"'] \\
\mathscr{A}(\mathbf{m}^\beta) \arrow[r, bend left, "\mathbb{F}_{\mathbf{m}^\beta}"] \arrow[r, bend right, "\mathbb{G}_{\mathbf{m}^\beta}"'] \arrow[r, phantom, "\underset{\lambda}{\Longrightarrow}"] & \mathscr{B}(\mathbf{m}^\beta)
\end{tikzcd}$$

7 Transition to Equivariant Correspondance

The eventual goal is constructing an $\mathscr{F}_{G,\mathbb{R}}$-pseudoalgebra out of an $\mathscr{P}_G$-pseudoalgebra, while extend it to a 2-functor. [8] provides a general construction for the non-equivariant case (or $G = \mathbf{1}$).

Given a $\mathscr{P}$-pseudoalgebra $\mathscr{A}$, define object map $\mathscr{B} : \mathscr{F} \to \mathbf{Cat}$ by $\mathscr{B}(\mathbf{n}) := \mathscr{A}^n$. For extending to a pseudofunctor (and restricting to functor on Π), the central concepts include:

(i) Substituting the vanishing coordinates of maps $\psi : \mathbf{n} \to \mathbf{m}$ in Π using the specified baseopint $\theta_0 : * \cong \mathscr{P}(0) \to \mathscr{A}$. In particular, since each $j \in \mathbf{m}$ has preimage being empty or a singleton, define $\psi : \mathscr{A}^n \to \mathscr{A}^m$ by

$$\psi(a_1, ..., a_n) := (a_{\psi^{-1}(1)}, ..., a_{\psi^{-1}(m)}), \tag{7.1}$$

with $a_{\psi^{-1}(j)} := *$ (the basepoint) if $\psi^{-1}(j) = \emptyset$. This defines $\mathscr{B}$ to be a functor with source Π.

(ii) Treating each canonical n-fold product $\phi_n : \mathbf{n} \to \mathbf{1}$ in $\mathscr{E}$ (with $\phi_n(j) = 1$ for all $j \neq 0$) using the n-fold product $e \times \mathscr{A} \hookrightarrow \mathscr{P}(n) \times \mathscr{A} \xrightarrow{\gamma} \mathscr{A}$.

(iii) Finally, show the decomposition laws, that maps in $\mathscr{E}$ are compositions of permutation and wedge of cannoical products (when fixing a specific order, it's unique), while maps in $\mathscr{F}$ decomposes into a projection and a map in $\mathscr{E}$. Using the laws to construct functors $\mathscr{A}^n \to \mathscr{A}^m$ using each map ψ.

The coherent natural isomorphisms for $\mathscr{P}$-pseudoalgebra provides the pseudofunctoriality modulo extensive diagram chasing.

Step (i) is generalizes naturally to the equivariant case, as each $\mathscr{P}_G$-pseudoalgebra $\mathscr{A}$ has a natural basepoint specified by $\theta_0 : * \cong \mathscr{P}_G(0) \to \mathscr{A}$, each $\mathscr{A}^{\mathbf{n}^\alpha} \cong \mathscr{A}^n$ nonequivariantly, and the maps in Π_G includes all nonequivariant ones, hence the same construction carries over.

For (ii), each canonical product $\phi_{\mathbf{n}^\alpha} : \mathbf{n}^\alpha \to \mathbf{1}$ has a choice given by $\alpha \times \mathscr{A}^n \hookrightarrow \mathscr{P}_G(n) \times \mathscr{A}^n \xrightarrow{\theta_n} \mathscr{A}$ (by treating), with some equivariance details we'll specify below.

For (iii), the main problem for equivariant case is: Not every map can be expressed as wedges of canonical product, as the preimage of an element may not necessarily be wedges of G-orbits. So, potentially some choices

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